
दही — विशिष्टि

(पहला पुनरीक्षण)

Dahi — Specification

(*First Revision*)

ICS 67.100.10

© BIS 2023



भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुर शाह ज़फर मार्ग, नई दिल्ली - 110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI - 110002
www.bis.gov.in www.standardsbis.in

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Dairy Products and Equipment Sectional Committee had been approved by the Food and Agriculture Division Council.

Due to good digestibility, large section of population in India consume fermented milk products specially *Dahi*. The quality of *Dahi* and yoghurt depends on the starter culture used, initial quality of milk and other ingredients added. To cover specification for fermented milk products, covering *Dahi* and yoghurt, IS 7035 was published in 1973 with the title 'Specification for Fermented Milk Products'. While revising the standard, the sectional committee decided to split the standard into two separate standards, one covering *Dahi* published as IS 9617 : 1980 'Specification for *Dahi*' and other covering yoghurt published as IS 12898 : 1989 'Dairy Products — Yoghurt — Specification', the way these terms are generally understood and these products are commonly marketed in the country.

IS 9617 : 1980 version incorporated a number of modifications, namely (a) range of acidity was given, (b) different types of milk were included, (c) cultures to be used were made explicit, and (d) packing clause was made more comprehensive.

This revision has been brought out to harmonize the standard with the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011, and the major changes include:

- a) Acidity requirement has been modified;
- b) Requirements for different types of flavoured *Dahi* in line with FSSAI Regulations have been included;
- c) Requirement of milk protein has been introduced; and
- d) Microbiological requirements have been aligned with *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011.

In the formulation of this standard, due consideration has been given to the provisions of the *Food Safety and Standards Act*, 2006 and the Rules and Regulations framed thereunder and the *Legal Metrology (Packaged Commodities) Rules*, 2011. However, this standard is subject to the restrictions imposed under these, wherever applicable.

The composition of the committee responsible for formulation of the standard is listed in Annex C.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

DAHI — SPECIFICATION

(First Revision)

1 SCOPE

1.1 This standard prescribes the types, requirements and the methods of sampling and test for *Dahi*. It covers *Dahi* obtained by lactic fermentation through the action of single or mixed strains of lactic acid bacteria with or without additions as specified under 5.2 and 5.3.

1.2 This standard does not include milk coagulated by the addition of acids, milk coagulating enzymes, etc.

2 REFERENCES

The standards listed in Annex A contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

3 TYPES

3.1 *Dahi* shall be of the following types:

- a) Plain *Dahi*; and
- b) Flavoured *Dahi*.

3.2 The culture for preparation of *Dahi* shall be any lactic acid bacteria, single or in combination.

4 INGREDIENTS

4.1 Raw Materials

The following are the essential raw materials for the manufacture of *Dahi*.

4.1.1 Milk and Milk Products

- a) Milk, as defined in IS 13688, pasteurized (see IS 13688) or boiled; and/or
- b) Concentrated milk and dried milk and other milk solids; and/or
- c) Cream, white/cooking butter, butter oil and anhydrous milk fat.

4.1.2 Lactic Acid Bacteria

4.2 Permitted Ingredients

Flavoured *Dahi* may contain the following optional ingredients:

- a) Sugar;
- b) Salt;
- c) Nutritive sweeteners other than sugar;
- d) Non-dairy ingredients such as fruits and vegetables and their products such as juices, purees, pulps, preparations and preserves derived therefrom, cereals and cereal products, coconut and coconut products, honey, chocolate, nuts, coffee, spices, condiments, culinary herbs and other harmless natural flavouring foods; and
- e) Prebiotics and Probiotics.

4.3 The flavoured *Dahi* may contain permitted food additives within the limits as specified under the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations, 2011*.

5 REQUIREMENTS

5.1 Description

Dahi, a fermented milk product, shall be obtained by fermentation of milk manufactured using other permitted ingredients, by the action of lactic acid bacteria. *Dahi* shall have a pleasing flavour bouquet resulting from the blend of a clean delicate somewhat aromatic odour and a clean acid taste. It shall not have off-odour, bitter and metallic taste. It shall be free from undesirable flavours such as [flat (lack of flavors), unclean, oxidized/rancid and yeasty]. Before shaking, *Dahi* shall be firm, solid and uniform with negligible whey separation. *Dahi* shall not have coarse texture and ropiness.

5.2 Flavoured *Dahi* shall contain a maximum of 50 percent (m/m) of permitted non-dairy ingredients including flavourings. The non-dairy ingredients can be mixed prior to or after fermentation.

NOTE — The term 'flavoured' covers sweetened, flavoured and fruit variants.

5.3 *Dahi* shall have the same minimum percentage of milk fat and milk solids-not-fat as that of the milk, from which it is prepared. Where *Dahi* is sold or offered for sale without any indication of class of

from mixed milk (*see* IS 13688) shall apply.

5.4 Flavoured *Dahi* shall be of three types based on the fat content as indicated in Table 1.

Table 1 Types of Flavoured *Dahi*
(Clause 5.4)

Sl No.	Parameter	Requirement			Method of Test, Ref to
		Flavoured <i>Dahi</i>	Flavoured Partly Skimmed <i>Dahi</i>	Flavoured Skimmed <i>Dahi</i>	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Milk Fat, percent, (<i>m/m</i>)	Not less than 3.0 and not more than 15.0	More than 0.5 and Less than 3.0	0.5, <i>Max</i>	IS 1224 (Part 1) or 5 of IS 1479 (Part 2)*
ii)	Milk solids-not-fat, minimum, percent, (<i>m/m</i>)	9.0	9.0	9.0	IS 12233
iii)	Added sugar, percent (<i>m/m</i>)				Annex B of IS 4079
	a) <i>Mishti Dahi</i>	15	15	15	
	b) Variants of Flavoured <i>Dahi</i> other than <i>Mishti Dahi</i>	10	10	10	

NOTE — In case of dispute, methods indicated by “*” shall be the referee method.

5.5 The product shall also conform to the requirements given in Table 2 and Table 3.

Table 2 Requirements for Plain *Dahi* and Flavoured *Dahi*
(Clause 5.5)

Sl No.	Characteristic	Requirement	Method of Test, Ref to
(1)	(2)	(3)	(4)
i)	Acidity, as lactic acid, percent by mass, <i>Min</i>	0.6	12 of IS 1479 (Part 1)
ii)	Milk protein, percent by mass, <i>Min</i>	3.0	IS 11917
iii)	Phosphatase test	Negative	IS 8479 (Part 1)

Table 3 Microbiological Requirements for Plain *Dahi* and Flavoured *Dahi*
(Clause 5.5)

SI No.	Characteristic	Requirement				Method of Test, Ref to
		Sampling Plan		Limit (cfu)		
		<i>n</i>	<i>c</i>	<i>m</i>	<i>M</i>	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Coliform count	5	2	10/g	1×10^2 /g	IS 5401 (Part 1 or Part 2)
ii)	<i>Staphylococcus aureus</i> (Coagulase positive)	5	2	10/g	1×10^2 /g	IS 5887 (Part 8/ Sec 1 or Sec 2)
iii)	Yeast and mold count	5	3	50/g	1×10^2 /g	IS 16069 (Part 1)
iv)	<i>Escherichia coli</i>	5	0	Absent/g	–	IS 5887 (Part 1)
v)	<i>Salmonella sp.</i>	5	0	Absent/25 g	–	IS 5887 (Part 3/ Sec 1)
vi)	<i>Listeria monocytogenes</i>	5	0	Absent/g	–	IS 14988 (Part 1)

NOTE — For sampling plan, *see* Annex B.

5.6 Hygienic Conditions

The product shall be processed, packed, stored and distributed under hygienic conditions (see IS 2491).

6 PACKING AND MARKING

6.1 Packing

The product shall be packed in clean and sound glass containers or inert plastic containers like PET and HDPE Jars or in a food grade flexible packaging material made from a film or combination of any of the substrates made of board, paper, polyethylene, polyester metalized film or aluminium foil in such a way so as to protect it from deterioration. In the case of the plastic material, only food grade plastic (see IS 10171) of minimum thickness 0.05 mm shall be used. The product shall be cooled before dispatch and maintained preferably at a temperature below 5 °C.

6.2 Marking

The following information shall be marked legibly and indelibly on each container:

- Name of the manufacturer, and trademark, if any;

- Name and type of the product (Plain *Dahi* or Flavoured *Dahi*);
- Sweetened or unsweetened;
- Net quantity;
- Permitted food additives, if added;
- Date of manufacture or packaging;
- Expiry/use by date;
- Batch or code number; and
- Any other requirements under the *Food Safety and Standards (Labelling and Display) Regulations, 2020* and the *Legal Metrology Act, 2009* and Rules framed thereunder.

6.2.1 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

7 SAMPLING

Representative samples of *Dahi* for testing conformity to this standard shall be drawn as described in IS 11546.

ANNEX A
(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
IS 1224 (Part 1) : 1977	Determination of fat by Gerber method: Part 1 Milk		<i>aureus</i> and other species), Section 2 Technique using rabbit plasma fibrinogen agar medium
IS 1479	Methods of test for dairy industry:		
(Part 1) : 2016	Rapid examination of milk (<i>first revision</i>);	IS 8479 (Part 1): 1977	Method for determination of phosphatase activity in milk and milk products: Part 1 Routine method
(Part 2) : 1961	Chemical analysis of milk		
2491 : 2013	Food hygiene — General principles — Code of practice (<i>third revision</i>)	IS 10171 : 1999	Guide on suitability of plastics for food packaging (<i>second revision</i>)
IS 4079 : 2023	Packed <i>Rasogolla</i> — Specification (<i>first revision</i>):	IS 11546 : 2012/ ISO 707 : 2008	Milk and milk products — Guidance on sampling (<i>second revision</i>)
(Part 1) : 2012/ISO 4832 : 2006	Colony count technique (<i>second revision</i>);		
(Part 2) : 2012/ISO 4831 : 2006	Most probable number technique (<i>second revision</i>)	IS 11917 : 2018	Milk and milk products — Determination of nitrogen content — Kjeldahl principle and crude protein calculation (<i>first revision</i>)
IS 5401	Microbiology of food and animal feeding stuffs — Horizontal method for the detection and enumeration of coliforms	IS 12233 : 2017	Milk, cream and evaporated milk — Determination of total solids content (reference method) (<i>second revision</i>)
IS 5887	Methods for detection of bacteria responsible for food poisoning:	IS 13688 : 2020	Packaged pasteurized milk — Specification (<i>second revision</i>)
(Part 1) : 1976	Isolation, identification and enumeration of <i>Escherichia coli</i> (<i>first revision</i>);	IS 14988 (Part 1) : 2020/ISO 11290-1 : 2017	Microbiology of food and feeding stuffs — Horizontal method for detection and enumeration of <i>Listeria monocytogenes</i> : Part 1 Detection method (<i>first revision</i>)
(Part 3/Sec 1) : 2020/ISO 6579-1 : 2017	Horizontal method for the detection, enumeration and serotyping of <i>Salmonella</i> , Section 1 Detection of <i>Salmonella</i> spp. (<i>third revision</i>);		
(Part 8/Sec 1) : 2002/ISO 6888-1 : 1999	Horizontal method for enumeration of coagulase-positive <i>Staphylococci</i> (<i>Staphylococcus aureus</i> and other species), Section 1 Technique using Baird-Parker agar medium;	IS 16069 (Part 1) : 2013/ISO 21527-1 : 2008	Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of yeasts and moulds: Part 1 Colony count technique in products with water activity greater than 0.95
(Part 8/Sec 2) : 2002/ISO 6888-2 : 1999	Horizontal method for enumeration of coagulase - positive (<i>Staphylococcus</i>		

ANNEX B
(Table 3)

SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

B-1 SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

The terms n , c , m and M used in this standard have the following meaning:

n = Number of units comprising a sample;
 c = Maximum allowable number of units having microbiological counts above m for 2-class

sampling plan and between m and M for 3-class sampling plan;
 m = Microbiological limit that separates unsatisfactory from satisfactory in a 2-class sampling plan or acceptable from satisfactory in a 3-class sampling plan; and
 M = Microbiological limit that separates unsatisfactory from satisfactory in a 3-class sampling plan.

B-2 INTERPRETATION OF RESULTS

<i>2-Class Sampling Plan (where n, c and m are specified)</i>	<i>3-Class Sampling Plan (where n, c, m and M are specified)</i>
1) Satisfactory, if all the values observed are $\leq m$	1) Satisfactory, if all the values observed are $\leq m$
2) Unsatisfactory, if one or more of the values observed are $> m$ or more than c values are $> m$	2) Acceptable, if a maximum of c values are between m and M and the rest of the values are observed as $\leq m$
	3) Unsatisfactory, if one or more of the values observed are $> M$ or more than c values are $> m$

ANNEX C

(Foreword)

COMMITTEE COMPOSITION

Dairy Products and Equipment Sectional Committee, FAD 19

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Research Institute, Karnal	DR DHEER SINGH (Chairperson) DR MANMOHAN SINGH CHAUHAN (Former Chairperson)
All India Food Processors Association, New Delhi	DR K. L. GABA SHRI VIJAY GAUR (<i>Alternate</i>)
Bihar State Cooperative Milk Producers' Federation Ltd, (COMPFED), Patna	SHRI SUSHIL KUMAR SHRI RUPESH RAJ (<i>Alternate</i>)
Centre for Analysis and Learning in Livestock and Food (CALF), Anand	DR RAJESH NAIR DR RAJEEV CHAWLA (<i>Alternate</i>)
Confederation of Indian Food Trade & Industry, New Delhi	MS VARSHA YADAV DR ANIRUDHA CHHONKAR (<i>Alternate</i>)
Confederation of Indian Industry, New Delhi	MS NEHA AGGARWAL MS ARTI GUPTA (<i>Alternate</i>)
Directorate of Marketing and Inspection, Faridabad	DR D. M. GOVIND REDDY SHRI RAHUL SINGH (<i>Alternate</i>)
Export Inspection Council of India, New Delhi	DR J. S. REDDY SHRI KUMAR NARENDER (<i>Alternate</i>)
Envirocare Labs Pvt Ltd, Thane	SHRI NILESH AMRITKAR MS PRITI AMRITKAR (<i>Alternate</i>)
Food Safety and Standards Authority of India, New Delhi	DR MONICA PUNIYA
Gujarat Cooperative Milk Marketing Federation Ltd, Anand	SHRI SAMEER SAXENA SHRI SAYAN BANERJEE (<i>Alternate</i>)
Indian Dairy Association, New Delhi	DR G. S. RAJORHIA DR SATISH KULKARNI (<i>Alternate</i>)
IDMC Ltd, Anand	SHRI DEVENDER GUPTA SHRI PRAKASH MAHESHWARI (<i>Alternate</i>)
Indian StainlessSteel Development Association, Gurgaon	SHRI ROHIT KUMAR SHRI RAJAT AGGARWAL (<i>Alternate</i>)
Jupitor Glass Works, New Delhi	SHRI KARAN NANGIA SHRI AMREEK SINGH PURI (<i>Alternate</i>)
Ministry of Fisheries, Animal Husbandry and Dairying, Department of Animal Husbandry and Dairying, New Delhi	SHRI GOUTAM KUMAR DEB SHRI AJIT KUMAR K. (<i>Alternate</i>)
Mother Dairy Fruit and Vegetable Ltd, New Delhi	MS NITA SEN SHRI SHAILENDER KUMAR (<i>Alternate</i>)

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Development Board, Anand	SHRI S. D. JAISINGHANI SHRI SURESH PAHADIA (<i>Alternate</i>)
National Dairy Research Institute, Karnal	DR VIVEK SHARMA DR RAJESH KUMAR BAJAJ (<i>Alternate</i>)
National Institute of Food Technology Entrepreneurship & Management (NIFTEM), Sonapat	DR P. K. NEMA
National Institute of Nutrition, Hyderabad	DR B. SANTOSH KUMAR DR SYLVIA FERNANDEZ RAO (<i>Alternate</i>)
Punjab State Coop. Milk Producers' Federation Ltd	DR SANJEEV KUMAR SHARMA
Rajasthan Co-op Dairy Federation (RCDF) Ltd, Jaipur	SHRI J. D. SINGH
Tamil Nadu Co-op Milk Producers' Federation Ltd, Chennai	SHRI S. R. SANKAR SHRI S. JEYACHANDRAN (<i>Alternate</i>)
Tetra Pak India Pvt Ltd, Pune	SHRI SHASHIKANT RAMNATH SURUSE SHRI SAMEER SINGH SUHAIL (<i>Alternate</i>)
Vimta Labs Ltd, Hyderabad	DR JAGADEESH KODALI DR MUNI NAGENDRA PRASAD POOLA (<i>Alternate</i>)
BIS Directorate General	SHRIMATI SUNEETI TOTEJA, SCIENTIST 'E'/DIRECTOR AND HEAD (FOOD AND AGRICULTURE) [(REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>))]

Member Secretary

DR BHAWANA
SCIENTIST 'D'/JOINT DIRECTOR
(FOOD AND AGRICULTURE), BIS

Panel for Revision of Indian Standards on Traditional Indian Dairy Products, FAD 19/Panel XIII

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Research Institute, Karnal	DR KAUSHIK KHAMRUI (Convenor)
Bihar State Milk Cooperative Federation Ltd, Patna	SHRI RUPESH RAJ
Bikanervala Foods Pvt Ltd	DR RAJESH GUPTA
Food Safety and Standards Authority of India, New Delhi	SHRI SUNIL BAKSHI
Gujarat Cooperative Milk Marketing Federation Ltd, Anand	SHRI SAYAN BANERJEE
Haldiram Sweets, New Delhi	MS RIKA RANJAN
In Personal Capacity (<i>House No. 616, Sector 2, Panckula – 134112</i>)	DR D.K. SHARMA

IS 9617 : 2023

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Development Board, Anand	SHRI ADITYA JAIN
Southern Regional Station, National Dairy Research Institute, Bengaluru	DR K. JAYARAJ RAO

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act, 2016* to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright

BIS has the copyright of all its publications. No part of these publications may be reproduced in any form without the prior permission in writing of BIS. This does not preclude the free use, in the course of implementing the standard, of necessary details, such as symbols and sizes, type or grade designations. Enquiries relating to copyright be addressed to the Head (Publication & Sales), BIS.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the website- www.bis.gov.in or www.standardsbis.

This Indian Standard has been developed from Doc No.:FAD 19 (18137).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002
Telephones: 2323 0131, 2323 3375, 2323 9402

Website: www.bis.gov.in

Regional Offices:

	Telephones
Central : 601/A, Konnectus Tower -1, 6 th Floor, DMRC Building, Bhavbhuti Marg, New Delhi 110002	{ 2323 7617
Eastern : 8 th Floor, Plot No 7/7 & 7/8, CP Block, Sector V, Salt Lake, Kolkata, West Bengal 700091	{ 2367 0012 2320 9474
Northern : Plot No. 4-A, Sector 27-B, Madhya Marg, Chandigarh 160019	{ 265 9930
Southern : C.I.T. Campus, IV Cross Road, Taramani, Chennai 600113	{ 2254 1442 2254 1216
Western : Plot No. E-9, Road No.-8, MIDC, Andheri (East), Mumbai 400093	{ 2821 8093

Branches : AHMEDABAD. BENGALURU. BHOPAL. BHUBANESHWAR. CHANDIGARH. CHENNAI. COIMBATORE. DEHRADUN. DELHI. FARIDABAD. GHAZIABAD. GUWAHATI. HIMACHAL PRADESH. HUBLI. HYDERABAD. JAIPUR. JAMMU & KASHMIR. JAMSHEDPUR. KOCHI. KOLKATA. LUCKNOW. MADURAI. MUMBAI. NAGPUR. NOIDA. PANIPAT. PATNA. PUNE. RAIPUR. RAJKOT. SURAT. VISAKHAPATNAM.